ON CREDIT

SCORING FOUNDATIONS IN THE AGE OF AI

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Preface

This book traces the mathematical lineage connecting Alan Turing's wartime cryptanalysis to modern credit scoring systems. It reveals how weight of evidence, developed to break enemy codes at Bletchley Park near London in the early 1940s, connects through maximum likelihood and logistic regression to neural networks, ensemble methods, and the explainability frameworks used in today's financial institutions.

The journey follows a unified thread: *the credit assignment problem*. Whether assigning credit to loan applicants, attributing predictions to features through SHAP values, or distributing error signals backward through neural network layers, the fundamental challenge remains the same: fairly allocating responsibility across contributing factors. In this sense, it is all about assigning credit, both literally and figuratively.

The additive log-odds framework underlying credit scorecards shares its mathematical DNA with neural networks, gradient boosting, and SHAP values. These methods are not separate paradigms but variations on a single mathematical theme rooted in additive log-odds and maximum likelihood. All of them trace back to the same principle Turing used at Bletchley Park: accumulating weights of evidence through additive scoring.

This book is not a manual of techniques but an exploration of the conceptual foundations that underlie credit risk modeling's past and future. It is written for data scientists and risk professionals working in financial services, as well as students and researchers interested in the foundations of credit scoring and modern banking. Rather than focusing on a specific programming language, this book emphasizes conceptual understanding and mathematical intuition, making the material accessible to both technical practitioners and senior risk leaders.

Each chapter builds on the previous one, connecting concepts, historical figures, and ideas across seemingly separate worlds, with dedicated bibliographies supporting focused study. Chapters 1 and 2 trace the statistical lineage, from Turing's wartime cryptanalysis through Fisher's maximum likelihood to the birth of logistic regression. Chapters 3 and 4 shift to the early foundations of AI and classical machine learning, revealing how these parallel traditions converged on the same mathematical principles. The book shows they were never truly separate.

It is the author's hope that this book serves as both a conceptual map and a practical reference for those seeking to understand credit scoring's place in the broader history of statistical thought.

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About the Author

Denis Burakov specializes in credit risk data science, with experience across banking and fintech in developed and emerging markets. He has held senior roles at Amazon, N26, KPMG, and Sberbank, spanning retail and corporate lending, decisioning, regulatory and managerial risk models, and fraud detection. His work bridges traditional statistical methods with modern AI approaches in financial services. He is based in Berlin, Germany.

About the Book Cover

The cover illustration is adapted from the title page of "Libro che tracta di mercatantie et usanze de paesi" (The book of commerce and customs of the countries, 1497) by Giorgio di Lorenzo Chiarini showing a Florentine bank, likely a branch of the famous Medici Bank.

This Renaissance-era woodcut depicts merchants engaged in financial transactions and record-keeping. The Medici Bank was one of the earliest institutions which innovated in banking by pioneering double-entry accounting and a decentralized branch system.

Contents

Introduction

Credit risk modeling occupies a special place in the landscape of applied statistics and machine learning. When we read machine learning textbooks or review standard examples, we encounter fraud detection use cases but rarely credit scoring. This absence is telling.

As Leo Breiman noted in [2], commenting on Richard Bolton and David Hand's research [1], statisticians' lack of adaptability led some problems to be ceded to computer scientists and engineers in the machine learning community. This helps explain why credit scoring remains absent from many modern machine learning texts.

Unlike many domains where academic researchers develop methods that practitioners later adopt as it happened with fraud detection, credit scoring evolved differently: from the ground up, by practitioners solving real lending problems. As Martha Poon observes [4]:

> "*As with other historical movements of statistics, techniques for treating the credit application problem probabilistically can be seen developing in the hands of practitioners working on the ground with a domain experience, as opposed to being developed and diffused by academic or professional statisticians.*"

This practitioner-driven evolution explains much about the field's character. Credit scoring developed its own vocabulary, its own standards of evidence, and its own mathematical culture, in parallel to, rather than downstream from, academic statistics. The Weight of Evidence (WOE) framework, the emphasis on interpretable scorecards, and the regulatory requirements for explainability of decisions all emerged from the practical demands of lending decisions, not from theoretical considerations alone.

Yet beneath this practical surface runs a deeper mathematical current. The same principles that Alan Turing used to break enemy codes in 1941 (additive logarithmic scoring and incremental estimation from partial evidence) resurface in modern credit models, neural networks, and explainability frameworks. This book traces that current, showing how ideas about probability and learning evolved from wartime cryptanalysis through classical statistics into the machine learning systems that now govern credit decisions.

Key Arguments of the Book

This book positions credit scoring as a window into the broader history of reasoning under uncertainty.

Credit scoring emerges as a specialized machine learning domain defined by:

- strict interpretability constraints,
- regulatory boundaries,
- unique historical lineage, and
- methodology that predates and anticipates modern machine learning.

In other words, credit scoring is not behind mainstream machine learning methods, it optimizes for different constraints. In credit scoring, unlike other domains, explainability is not an aesthetic preference but a legal and operational requirement.

Yet this is often misunderstood. The European Central Bank's revised Guide on Internal Models [3] states: "For example, generalised linear models (such as linear or logistic regressions) are not deemed to be ML techniques for the purposes of this guide." This artificial separation obscures the shared foundations between logistic regression and modern classification algorithms.

These different constraints explain the parallel paths: credit scoring and AI grew together but separately, credit scoring evolving within the institutional constraints of banking while machine learning evolved through algorithmic experimentation.

What Influenced This Book

Several works deeply influenced this book's approach. Dan Jurafsky and James Martin's *Speech and Language Processing* inspired the historical narrative structure, while Alex K. Gold's *DevOps for Data Science*, Simone Scardapane's *Alice's Adventures in a Differentiable Wonderland*, and Andrew Trask's *Grokking Deep Learning* shaped the technical exposition. For the Turing material, I drew on Jack Copeland's *The Turing Guide* and Dermot Turing's *The Codebreakers of Bletchley Park*. Joan Box's *R.A. Fisher: The Life of a Scientist* provided essential context for Fisher's work. I.J. Good's writings (consulted at Berlin's State Library at Unter den Linden), and Yudi Pawitan's *In All Likelihood* informed the connection between weight of evidence and maximum likelihood. David Spiegelhalter's books, particularly *The Art of Statistics* and *The Art of Uncertainty*, greatly influenced my writing style. A visit to Bletchley Park in autumn 2024 enriched my treatment of Alan Turing's contributions.

The Book's Scope and Focus

This book traces a specific intellectual lineage: from Turing's weight of evidence through Fisher's maximum likelihood theory to machine learning and explainable AI. This book does not attempt to lay out a comprehensive narrative of how statistics and machine learning have evolved together. Instead, it follows a single thread, *the credit assignment problem*, showing how the challenge of fairly allocating credit (whether to loan applicants, model features, or neural network layers) connects seemingly disparate methods.

The narrative focuses on five core techniques that dominate modern credit scoring: logistic regression, neural networks, decision trees, ensemble methods (random forests and gradient boosting), and SHAP values. These methods share a common mathematical foundation in additive log-odds and maximum likelihood, making them natural companions in a unified treatment.

Our focus remains on the methods most central to credit risk modeling and the statistical principles that unite them: principles that, remarkably, trace back to a mathematician working in a secret facility during World War II.

What to Expect from Each Chapter

The journey in Chapter 1 begins in 1941 at Bletchley Park, where Alan Turing transformed wartime codebreaking into arithmetic. He took odds, logged them, and added tiny scores (decibans) until a hypothesis outweighed its rivals. Eight decades later, banks do the same thing every time you apply for credit. This chapter follows that thread from Turing's factor principle to information theory's view of evidence, and finally to credit scoring, showing how the same mathematics powers machines that talk, systems that learn, and the scorecards that decide our loans.

Chapter 2 traces the evolution from Fisher's maximum likelihood to modern logistic regression. We explore how Fisher's 1922 binomial example evolved through Berkson's practical simplification, examine parameter estimation from closed-form solutions to iterative optimization, and demonstrate the mathematical equivalence between weight of evidence transformations and logistic regression coefficients. The chapter extends these ideas to multinomial outcomes, showing how Fisher's likelihood principle became one of the most widely used algorithms in data science.

Unlike the previous chapters' linear progression, Chapter 3 weaves together multiple historical threads to tell a more complex story. We trace the forgotten origins of neural networks in the 1940s and 1950s, revealing how Turing's ideas about learning machines preceded the famous Perceptron. The chapter explores the AI renaissance with backpropagation and multi-layer perceptrons, showing how credit risk modeling evolved on a parallel track to mainstream AI.

Chapter 4 traces the evolution from decision trees to modern ensemble methods, highlighting Leo Breiman and Jerome Friedman's contributions ranging from bagging through random forests to gradient boosting. The chapter also introduces Lloyd Shapley, whose 1950s game theory solution to fair credit allocation now powers SHAP, the dominant framework for explaining complex machine learning models.

We examine both prediction and explanation: beginning with the mechanics of decision trees, tracing their evolution into ensembles, and connecting maximum likelihood to gradient boosting. These foundations set the stage for understanding SHAP's role in credit risk modeling, bridging Breiman's "two cultures" of statistics.

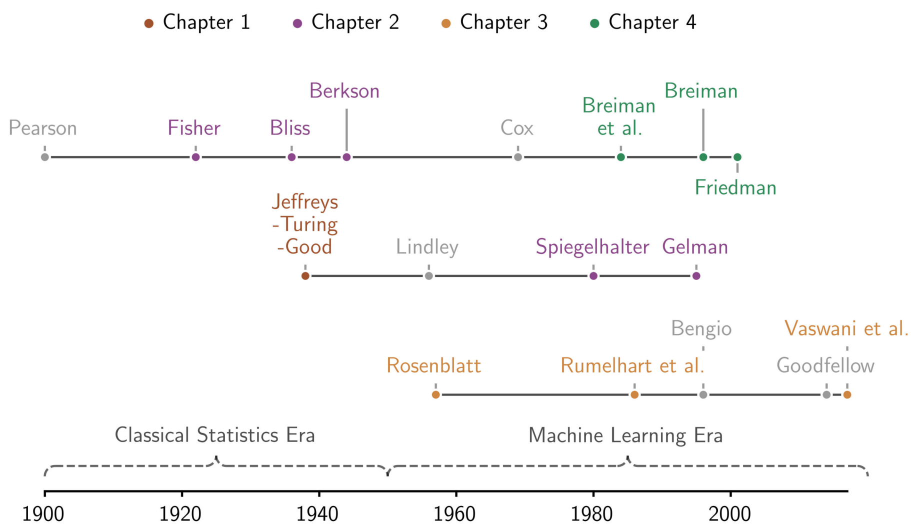

Figure 0.1: An opinionated intellectual lineage connecting key figures across the statistical, Bayesian, and machine learning traditions, showing which chapters cover their contributions. This timeline reflects the book's specific narrative focus and is not intended as a comprehensive historical account.

The journey from Turing's decibans to SHAP values spans eight decades, multiple disciplines, and two world-changing technologies: computers and artificial intelligence. Yet the mathematical core remains remarkably stable. Whether breaking codes, fitting logistic regressions, training neural networks, or explaining gradient boosting predictions, we return again and again to the same fundamental operation: adding up log-odds, accumulating evidence, and fairly attributing credit.

This book shows how that simple idea — transforming multiplication into addition through logarithms — became the foundation for modern credit risk modeling.

My sincere thanks go to Naeem Siddiqi for the foreword and to Agus Sudjianto for the epilogue to this book. Naeem's work on credit scoring introduced me to the field and continues to inspire practitioners worldwide, while Agus served as an invaluable mentor and advisor in advanced credit risk modeling. I also thank Paul Edwards, Andrija Djurovic, Valeriy Manokhin, Guillermo Navas-Palencia, Fabiana

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